

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:06-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

06-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-8(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

For More Material Join: @JEEAdvanced_2024

PHYSICS

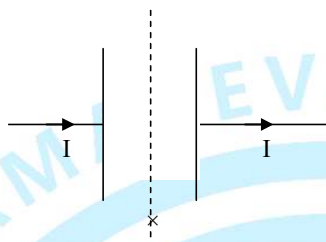
MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

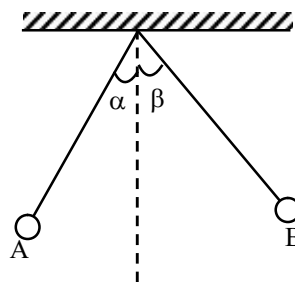
1. A circular parallel plate capacitor of radius R and distance d between the plate is given. A capacitor is being charged with a current I flowing through the wire. Neglect fringing effect.



What is the rate of change of electric flux through plane in middle of capacitor with respect to time (i.e. $\frac{d\phi}{dt}$) –

- A) $\frac{2I}{\epsilon_0}$ B) $\frac{I}{\epsilon_0}$ C) $\frac{4I}{\epsilon_0}$ D) $\frac{6I}{\epsilon_0}$
2. A particle is moving in a circular path. The acceleration and momentum of the particle at a certain moment are $\vec{a} = (4\hat{i} + 3\hat{j}) \text{ m/s}^2$ and $\vec{p} = (8\hat{i} - 6\hat{j}) \text{ kg-m/s}$. The motion of the particle at that instant is
- A) Uniform circular motion
B) accelerated circular motion
C) decelerated circular motion
D) We cannot say anything with $\vec{a}$ and $\vec{p}$ only
3. A block starts moving up a fixed inclined plane of inclination 60° with a velocity of 20 m/s and stops after 2 sec . The approximate value of coefficient of friction is ($g = 10 \text{ m/s}^2$)
- A) 3 B) 3.3 C) 0.27 D) 0.33

4. A ball of mass m approaches a wall of mass M ($\gg m$) with speed 4 m/s along the normal to the wall. The speed of wall is 1 m/s towards the ball. The speed of the ball after an elastic collision with the wall is -
- A) 5 m/s away from the wall B) 9 m/s away from the wall
C) 3 m/s away from the wall D) 6 m/s away from the wall
5. The height at which the acceleration due to gravity becomes $\frac{g}{9}$ (where g = the acceleration due to gravity on the surface of the earth) in terms of R , the radius of the earth, is
- A) $2R$ B) $\frac{R}{\sqrt{2}}$ C) $R/2$ D) $\sqrt{2}R$
6. An ideal gas is expanded so that amount of heat given is equal to the decrease in internal energy. The gas undergoes the process $TV^{1/5} = \text{constant}$. The adiabatic compressibility of gas when pressure is P , is -
- A) $\frac{7}{5P}$ B) $\frac{5}{7P}$ C) $\frac{2}{5P}$ D) $\frac{7}{3P}$
7. An ideal gas is held in a container of volume V at pressure P . The average speed of a gas molecule under these conditions is v . If now the volume and pressure are changed to $2V$ and $2P$, the average speed of a molecule will be
- A) $1/2 v$ B) v C) $2v$ D) $4v$
8. A wire is 4 m long and has a mass 0.2 kg . The wire is kept horizontally. A transverse pulse is generated by plucking one end of the taut (tight) wire. The pulse makes four trips back and forth along the cord in 0.8 sec . The tension in the cord will be -
- A) 80 N B) 160 N C) 240 N D) 320 N
9. Two identical simple pendulums A and B are fixed at same point. They are displaced by very small angles α and β ($\beta > \alpha$) and released from rest. Find the time after which B reaches its initial position for the first time. Collisions are elastic and length of strings is λ .



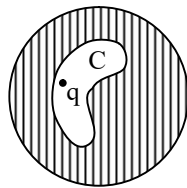
A) $\pi\sqrt{\frac{\ell}{g}}$

B) $2\pi\sqrt{\frac{\ell}{g}}$

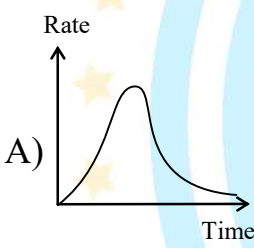
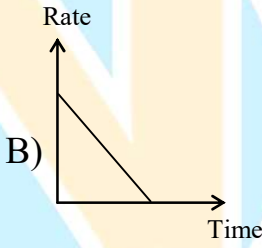
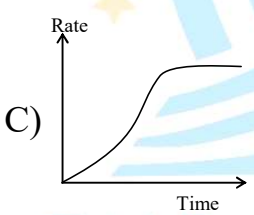
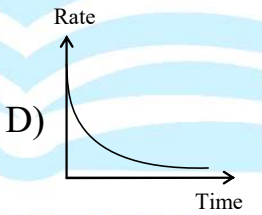
C) $\frac{\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$

D) $\frac{2\pi\beta}{\alpha}\sqrt{\frac{\ell}{g}}$

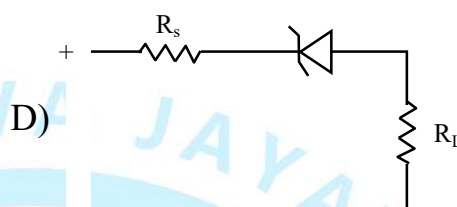
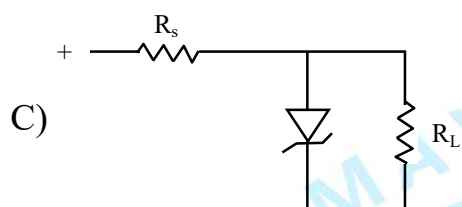
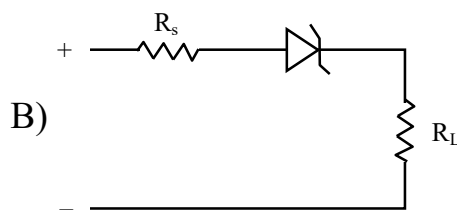
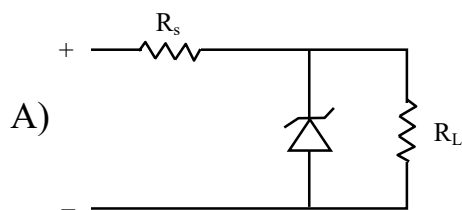
10. The figure shows a charge q placed inside a cavity C in an uncharged conductor. Now if an external electric field is switched on then :



- A) only induced charge on outer surface will redistribute.
- B) only induced charge on inner surface will redistribute
- C) Both induced charge on outer and inner surface will redistribute.
- D) force on charge q placed inside the cavity will change
11. STATEMENT – 1
- If potential difference between two points is zero and the resistance between the same two points is also zero, then current may flow between those two points
- STATEMENT – 2
- Kirchhoff's 1st law is conservation of charge.
- A) Statement – 1 is True. Statement – 2 is True; Statement – 2 is a correct explanation for Statement – 1.
- B) Statement – 1 is True, Statement – 2 is True; Statement – 2 is not a correct explanation for Statement – 1.
- C) Statement – 1 is True, Statement is False.
- D) Statement – 1 is False, Statement is True.

12. An α particle is moving along a circle of radius R with a constant angular velocity ω . Point A lies in the same plane at a distance $2R$ from the centre. Point A records magnetic field produced by α particle. If the minimum time interval between two successive times at which A records zero magnetic field is 't', then the angular speed ω is :
- A) $\frac{2\pi}{t}$ B) $\frac{2\pi}{3t}$ C) $\frac{\pi}{3t}$ D) $\frac{\pi}{t}$
13. Curie temperature is the temperature above which -
- A) a ferromagnetic material behaves like paramagnetic
B) a paramagnetic material behaves like diamagnetic
C) a ferromagnetic material behaves like diamagnetic
D) a paramagnetic material behaves like ferromagnetic
14. In an LR circuit connected to a battery, the time rate of change of energy stored in the inductor is plotted against time during the growth of current in the circuit. Which of the following best represents the resulting curve?
- A)  B) 
- C)  D) 
15. When an object is at distance x and y from a lens, a real image and a virtual image is formed respectively having same magnification. The focal length of the lens is -
- A) $\frac{x+y}{2}$ B) $x - y$ C) $\sqrt{xy}$ D) $x + y$
16. If the zero of the Vernier lies on the right hand side of zero of the main scale and fourth division coincide with the main scale division when the jaws are in contact, the zero correction will be ___ (assume standard Vernier Calipers)
- A) $+ 0.04$ cm B) $+ 0.06$ cm C) -0.04 cm D) -0.06 cm

17. A zener diode is to be used as a voltage regulator. Identify the correct set up –



18. The graph of $\log\left(\frac{R}{R_0}\right)$ versus $\log A$ (R = radius of a nucleus and A = mass number) is -

- A) a circle B) an ellipse C) a parabola D) a straight line

19. Match the entries of Column I with those in Column II

Column I		Column II	
(A)	Mass of products less than mass of reactants	(P)	Most stable
(B)	Iron nucleus	(Q)	Total internal reflection take place
(C)	Angle of contact is obtuse	(R)	Nuclear fission
(D)	Incidence angle more than critical angle	(S)	Stronger cohesive forces

A) A – R; B – P; C – Q ; D – S

B) A – R; B – P; C – S ; D – Q

C) A – Q; B – S; C – R ; D – P

D) A – Q; B – S; C – P ; D – R

20. In a YDSE experiment, I_0 is given to be the intensity of the central bright fringe & β is the fringe width. Then, at a distance y from central bright fringe, the intensity will be -
(y is very small)

A) $I_0 \cos\left(\frac{\pi y}{\beta}\right)$ B) $I_0 \cos^2\left(\frac{\pi y}{\beta}\right)$ C) $I_0 \cos\left(\frac{2\pi y}{\beta}\right)$ D) $I_0 \cos^2\left(\frac{\pi y}{2\beta}\right)$

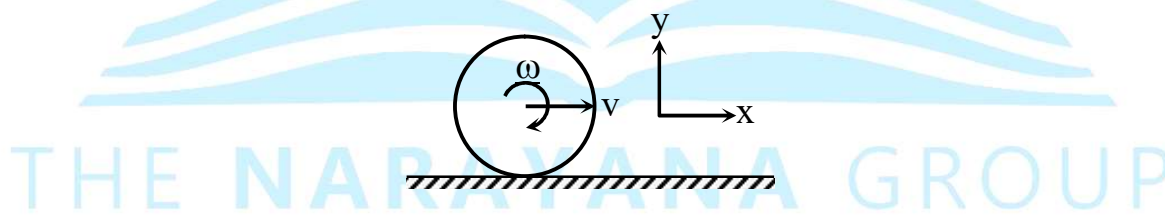
SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

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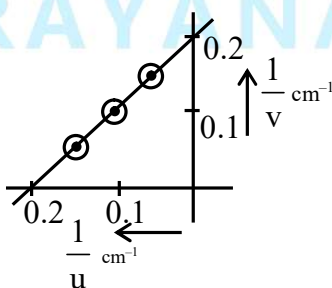
Marking scheme: +4 for correct answer, -1 in all other cases.

21. A particle of mass 10^{-2} kg is moving along the positive x-axis under the influence of a force $F(x) = -K/(2x^2)$ where $K = 10^{-2} \text{ Nm}^2$. At time $t = 0$ it is at $x = 1.0$ m and its velocity is $v = 0$. Find its velocity when it reaches $x = 0.50$ m. (in m/s)
22. A uniform ball of radius $R = 10$ cm rolls without slipping between two rails such that the horizontal distance is $d = 16$ cm between two contact points of the rail to the ball. If the angular velocity is 5 rad/s, then find the velocity of centre of mass of the ball in cm/s.
23. A disc of radius '5cm' rolls on a horizontal surface with linear velocity $v = 1 \hat{i}$ m/s and angular velocity 50 rad/sec. Height of particle from ground on rim of disc which has velocity in vertical direction is (in cm) -



24. A liquid flows out drop by drop from a vessel through a vertical tube with an internal diameter of 2 mm, then the total number of drops that flows out during 10 grams of the liquid flow out: [Assume that the diameter of the neck of a drop at the moment it breaks away is equal to the internal diameter of tube and surface tension is 0.02 N/m, $g = 9.8 \text{ m/s}^2$]

25. A tuning fork is in unison with a sonometer wire vibrating in its fourth overtone. Mass hanged with wire is 9 kg. When additional mass is hanged wire vibrates in unison with tuning fork in its 3rd harmonic. Additional mass hanged in kg is.
26. Two conducting concentric spherical shells are present. If the electric potential at the centre is 2000 V and the electric potential of the outer shell is 1500 V. then the potential of the inner shell in volts is
27. A series LCR circuit containing a resistance of 120Ω has angular resonance frequency $4 \times 10^5 \text{ rads}^{-1}$. At resonance the voltage across resistance and inductance are 60 V & 40 V respectively. At what frequency the current in the circuit lags the voltage by 45° . Give answer in $\text{-----} \times 10^5 \text{ rad/sec}$.
28. The energy stored in a 90 cm length of laser beam operating at 10 mW is $X \times 10^{-11} \text{ J}$. Find the value of X.
29. A thin prism P_1 with angle 4° and made from glass of refractive index 1.54 is combined with another thin prism P_2 made from glass of refractive index 1.72 to produce dispersion without deviation. The angle of prism in degrees is
30. An experiment with convex lens gives certain result which is represented by a student in the shown graph. The power of the lens in diopters is



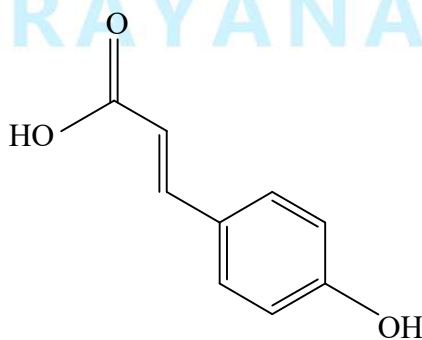
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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31. The element with $[\text{Xe}]4f^{14}5d^16s^2$ belongs to
- A) 7th period and Group 4 of the modern periodic table
 B) 6th period and Group 3 of the modern periodic table
 C) 5th period and Group 3 of the modern periodic table
 D) 6th period and Group 5 of the modern periodic table
32. The relative stability of +1 oxidation state of group 13 elements follows the order:
- A) $\text{Al} < \text{Ga} < \text{Tl} < \text{In}$ B) $\text{Ga} < \text{Al} < \text{In} < \text{Tl}$
 C) $\text{Tl} < \text{In} < \text{Ga} < \text{Al}$ D) $\text{Al} < \text{Ga} < \text{In} < \text{Tl}$
33. Assertion (A) : SO_2 turns acidified $\text{K}_2\text{Cr}_2\text{O}_7$ green.
 Reason (R) : $\text{K}_2\text{Cr}_2\text{O}_7$ is reduced to Cr^{3+} (green) salt by SO_2 .
- A) Both A and R are true and R is the correct explanation of A
 B) Both A and R are true but R is not the correct explanation of A
 C) A is true but R is false
 D) A is false but R is true
34. Product (X and Y) of the following reactions (1 and 2) are
- 1) $2\text{NaOH} + \text{Cl}_2 \rightarrow \text{NaCl} + \text{X} + \text{H}_2\text{O}$
(Cold and dilute)
- 2) $6\text{NaOH} + 3\text{Cl}_2 \rightarrow \text{NaCl} + \text{Y} + \text{H}_2\text{O}$
(Hot and conc.)
- A) $\text{X} = \text{NaClO}_3$ and $\text{Y} = \text{NaOCl}$ B) $\text{X} = \text{NaOCl}$ and $\text{Y} = \text{NaClO}_3$
 C) $\text{X} = \text{NaHClO}_3$ and $\text{Y} = \text{NaOCl}$ D) $\text{X} = \text{NaClO}_3$ and $\text{Y} = \text{NaHClO}_3$
35. Arrange $\text{Ce}^{+3}, \text{La}^{+3}, \text{Pm}^{+3}, \text{Yb}^{+3}$ in increasing order of their ionic radii
- A) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{Ce}^{+3} < \text{La}^{+3}$ B) $\text{Ce}^{+3} < \text{Yb}^{+3} < \text{Pm}^{+3} < \text{La}^{+3}$
 C) $\text{Yb}^{+3} < \text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3}$ D) $\text{Pm}^{+3} < \text{La}^{+3} < \text{Ce}^{+3} < \text{Yb}^{+3}$

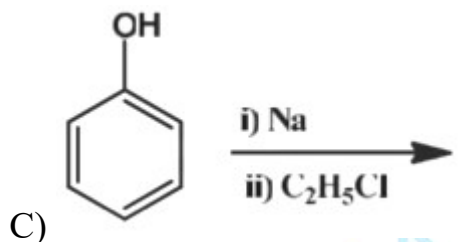
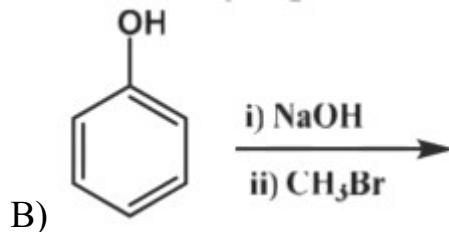
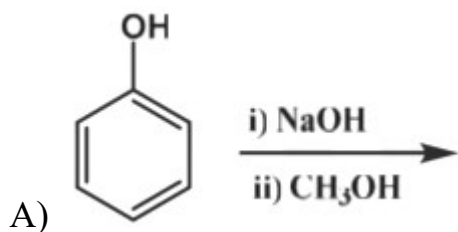
36. Which one of the following does not contain electrons in 5f subshell?
A) $\text{Lr}(Z=103)$ B) $\text{U}(Z=92)$ C) $\text{Cm}(Z=96)$ D) $\text{Th}(Z=90)$
37. Consider octahedral complex of Ca^{2+} with EDTA^{4-} .
 x = co-ordination number of complex
 y = Total number of O – donors. Find $(x+y)$?
A) 13 B) 14 C) 10 D) 11
38. In which of the following species maximum atoms can lie in same plane?
A) PCl_5 B) AsH_4^+ C) XeF_4 D) XeO_2F_2
39. Which of following involves maximum C – O bond length:
A) $\text{Ni}(\text{CO})_4$ B) $[\text{Fe}(\text{CO})_5]$ C) $[\text{Mn}(\text{CO})_6]^+$ D) $[\text{V}(\text{CO})_6]^-$
40. Assertion (A) : Coloured cations can be identified by borax – bead test
Reason (R) : Transparent bead ($\text{NaBO}_2 + \text{B}_2\text{O}_3$) forms coloured bead with coloured cation.
A) Both A and R are true and R is the correct explanation of A
B) Both A and R are true but R is not the correct explanation of A
C) A is true but R is false
D) A is false but R is true
41. Para-coumaric acid, an antioxidant in coffee, is treated with aqueous solution of bromine. What is the maximum number of bromine atoms that can be incorporated into a molecule of para-coumaric acid?



para – coumaric acid

- A) 1 B) 2 C) 3 D) 4

42. Which of the following reaction produces anisole as a major product?



D) All

43. The oxidation of toluene with hot KMnO_4 gives

A) Benzoic acid

B) Benzaldehyde

C) Benzene

D) Benzyl alcohol

44. Primary, secondary and tertiary amines can be separated by

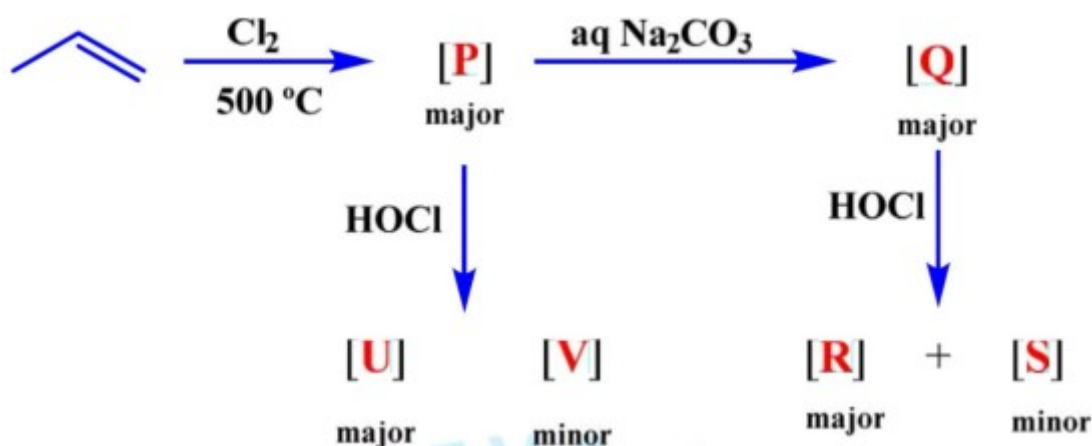
A) Hinsberg's test

B) HNO_2 test

C) Carbylamine test

D) Lucas reagent

45. Identify the **CORRECT** option(s) among the following ?



A) [R] & [S] on treatment with excess of dil NaOH gives same compound

B) Position of double bond in [P] & [Q] in same as per IUPAC

C) [U] & [V] on treatment with excess of dilute NaOH gives different product as [R] & [S] on same treatment

D) [U] & [R] are constitutional isomers

46. A 10cm^3 sample of an unknown gaseous hydrocarbon was mixed with 70cm^3 of oxygen and the mixture was set on fire by means of an electric spark. When the reaction was over and water vapours were liquefied, the final volume of gases decreased to 65cm^3 . This mixture then reacted with a potassium hydroxide solution and the volume of gases decreased to 45cm^3 . Find the molar mass of hydrocarbon in gm/mol, if volumes of gases were measured at standard temperature and pressure (STP) conditions ?

A) 26 B) 36 C) 16 D) 46

47. Radial wave function for 2s orbital of hydrogen – like atoms is given as

$$\psi = 2^{-3/2} \left[\frac{z}{a_0} \right]^2 \left[2 - \frac{zr}{a_0} \right] e^{-\frac{zr}{2a_0}}$$

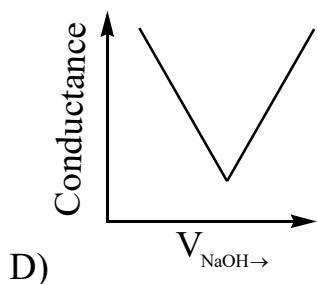
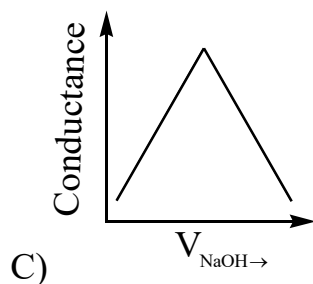
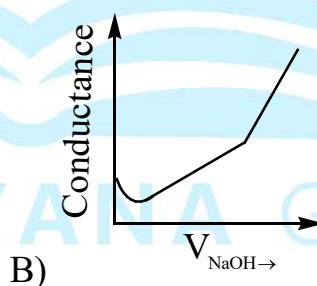
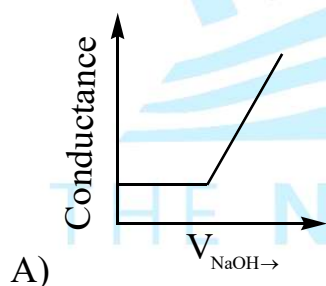
The radial probability ($4\pi r^2 \psi^2$) for an electron in He^+ at $r = a_0$ is

A) $\left[\frac{1}{a_0} \right]^2$ B) $2^{-3/2} \left[\frac{1}{a_0} \right]^2$ C) Zero D) $\left[\frac{1}{a_0} \right]^3$

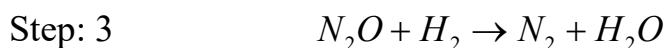
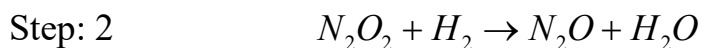
48. Benzene and toluene are nearly ideal solutions. If at a temperature $P_{\text{Benzene}}^0 = 120\text{mmHg}$ and $P_{\text{Toluene}}^0 = 50\text{mmHg}$. Then what will be the mole fraction of Toluene in vapour phase of this solution if mole fraction of Benzene in liquid phase of solution is 0.6.

A) 0.70 B) 0.6 C) 0.4 D) 0.22

49. Choose the correct representation of conductometric titration of benzoic acid vs sodium hydroxide.



50. Consider the reaction $2H_2(g) + 2NO(g) \rightarrow N_2(g) + 2H_2O(g)$. The rate law for this reaction is: $\text{rate} = k[H_2][NO]^2$ under what conditions could these steps represents the mechanism?



- A) These steps cannot be the mechanism under any circumstances
 B) These steps cannot be the mechanism if Step 1 is slow step
 C) These steps cannot be the mechanism if Step 2 is slow step
 D) These steps cannot be the mechanism if Step 3 is slow step

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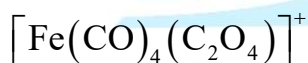
51. How many of these elements have lower electron affinity than fluorine?

Cl, S, O, N, P, Br, I, C

52.

Complex

Spin-only magnetic moment



a



b



c



d

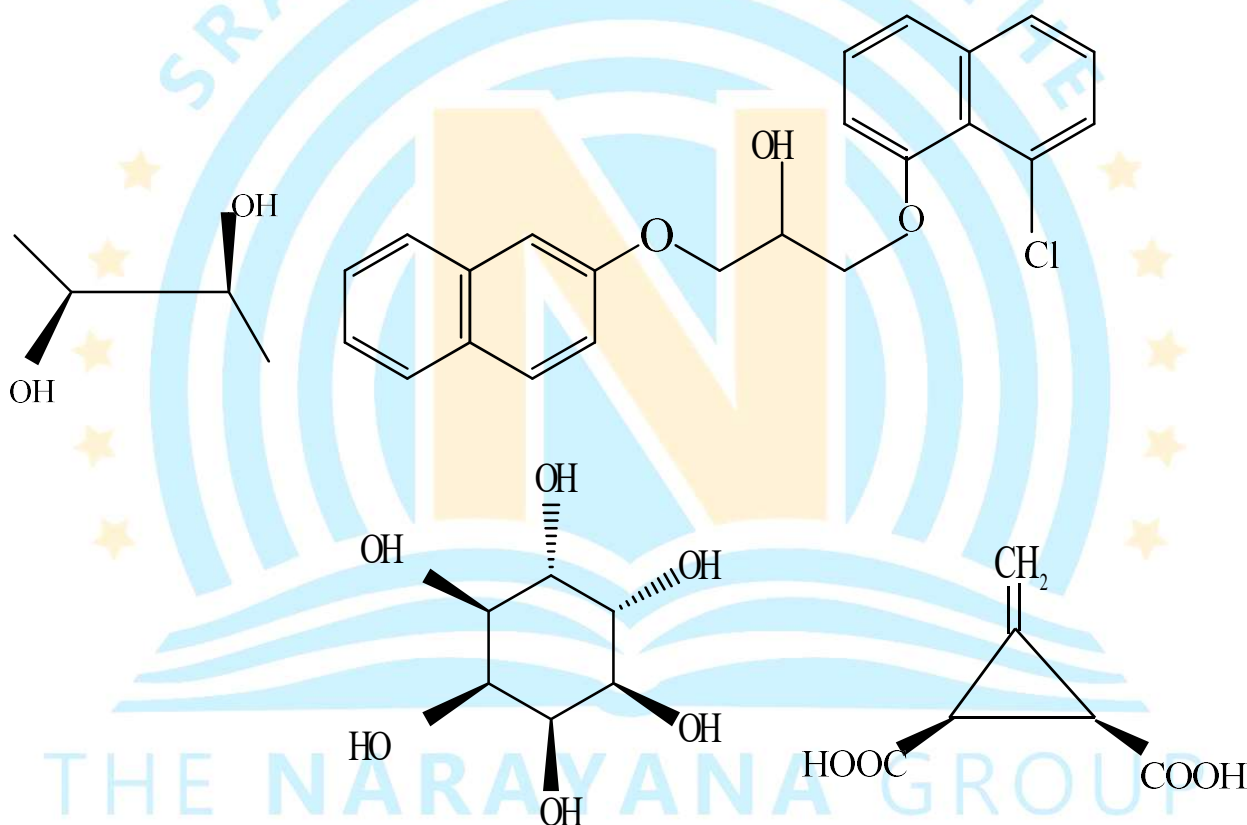
Then (a+b+c+d) = _____

(Given : $\sqrt{3} = 1.7$; $\sqrt{2} = 1.4$; $\sqrt{5} = 2.2$; $\sqrt{15} = 3.8$; $\sqrt{35} = 5.9$; $\sqrt{6} = 2.4$)

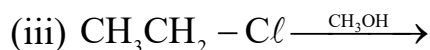
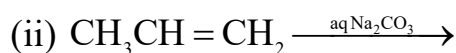
53. The number of correct statements is /are :

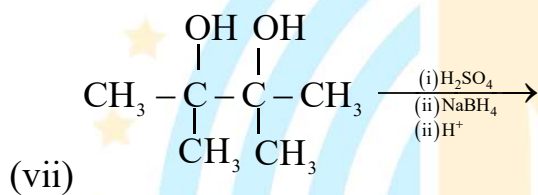
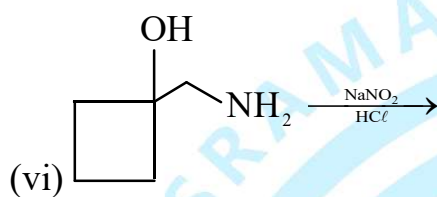
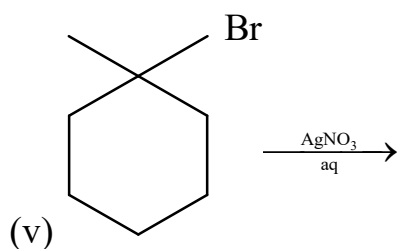
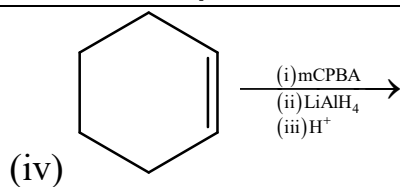
- (i) In manganate and permanganate ions. The π - bonding takes place by overlap of p-orbitals of oxygen and d-orbitals of manganese
- (ii) Manganate ion is green in colour and permanganate ion in purple in colour
- (iii) Manganate and permanganate ions are paramagnetic
- (iv) Manganate and permanganate ions are tetrahedral

54. The total number of chiral compound/s from the following is _____

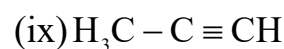
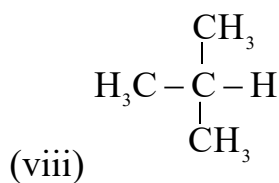
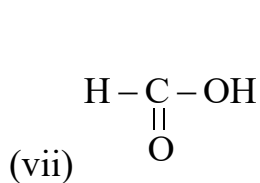
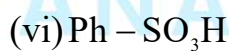
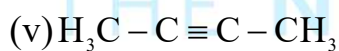
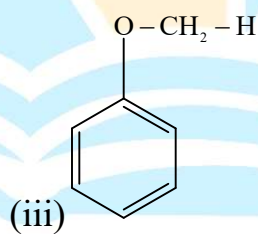
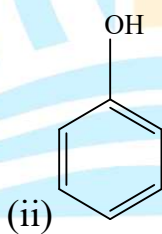
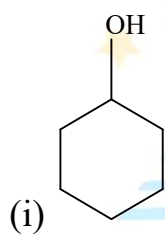


55. How many of the following reactions will give an alcohol as the major product ?

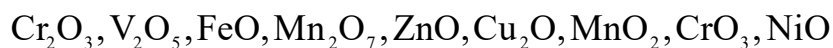




56. Number of compounds which can evolve H_2 on reaction with Na metal.



57. Among the following oxides



If x = number of basic oxides

y = number of acidic oxides

z = number of amphoteric oxides

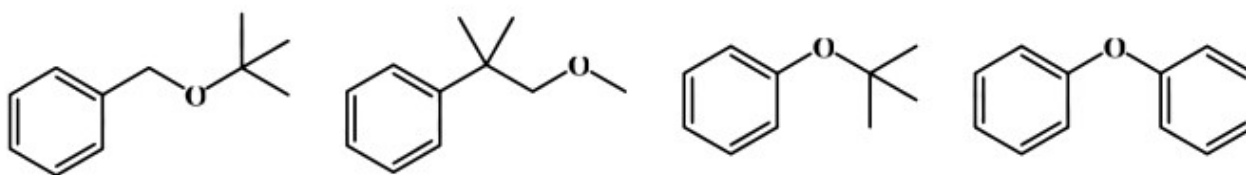
then the value of $\frac{(x+y)}{z}$ is

58. Find the solubility of $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{ClO}_4$.

If the $(\lambda_m^0)[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+ = 50\Omega^{-1}\text{cm}^2\text{mol}^{-1}$ $(\lambda_m^0)\text{ClO}_4^- = 70\Omega^{-1}\text{cm}^2\text{mol}^{-1}$ and the measured resistance was 33.5Ω in a cell with cell constant of 0.20cm^{-1} ?

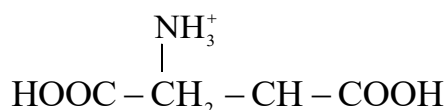
Represent your answer in milli mol/litre WITH NEAREST INTEGER

59. How many of following ethers are difficult to make or cannot be synthesized as a major product by S_N2 reaction?



60. At what pH given molecules doesnot migrate towards any electrode when electric field is applied. (give answer in nearest integer.)

$$(p^{ka_1} = 2; p^{ka_2} = 3.90; p^{ka_3} = 10.0)$$



MATHEMATICS

MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. Let z_1, z_2, z_3 be three distinct complex numbers lying in a circle with a centre at the origin such that $z_1 + z_2 z_3, z_2 + z_3 z_1$ and $z_3 + z_1 z_2$ are real numbers, then $z_1 z_2 z_3$ equals to
- A) -1 B) 0 C) 1 D) None of these
62. The number of irrational solutions of the equation $\sqrt{x^2 + \sqrt{x^2 + 11}} + \sqrt{x^2 - \sqrt{x^2 + 11}} = 4$ is
- A) 0 B) 2 C) 4 D) 11
63. If $H_1, H_2, \dots, H_n$ are n harmonic means between a and b , ($b \neq a$), then the value $\frac{H_1 + a}{H_1 - a} + \frac{H_n + b}{H_n - b}$ is equal to
- A) $n+1$ B) $n-1$ C) $2n$ D) $2n+3$
64. Let $H_n = 1 + \frac{1}{2} + \dots + \frac{1}{n}$ then the sum to n terms of the series $\frac{1^2}{1^3} + \frac{1^2 + 2^2}{1^3 + 2^3} + \frac{1^2 + 2^2 + 3^2}{1^3 + 2^3 + 3^3} + \dots$ is
- A) $\frac{4}{3}H_n - 1$ B) $\frac{4}{3}H_n + \frac{1}{n}$ C) $\frac{4}{3}H_n$ D) $\frac{4}{3}H_n - \frac{2}{3} \frac{n}{n+1}$
65. If $a = \log_{12} 18, b = \log_{24} 54$ then the value of $ab + 5(a - b)$ is
- A) 0 B) 4 C) 1 D) 6
66. The number of ordered pairs $(m, n), m, n \in \{1, 2, \dots, 100\}$ such that $7^m + 7^n$ is divided by 5 is
- A) 1250 B) 2000 C) 2500 D) 5000
67. The number of irrational terms in the expansion of $(5^{1/6} + 2^{1/8})^{100}$ is
- A) 96 B) 97 C) 98 D) 99
68. If A, B and C are three sets such that $A \cap B = A \cap C$ and $A \cup B = A \cup C$, then
- A) $A = C$ B) $B = C$ C) $A \cap B = \phi$ D) $A = B$

69. Two $n \times n$ square matrices A and B are said to be similar if there exists a non-singular matrix P such that $P^{-1}AP = B$. If A and B are similar and B and C are similar, then
- A) AB and BC are similar B) A and C are similar
C) $A+C$ and B are similar D) None of these
17. A letter is taken at random from the letters of the word 'STATICSTICS' and another letter is taken at random from the letters of the word 'ASSISTANT'. The probability that they are the same letter is
- A) $\frac{1}{45}$ B) $\frac{13}{90}$ C) $\frac{19}{90}$ D) $\frac{3}{16}$
71. $\sin 47^\circ + \sin 61^\circ - \sin 11^\circ - \sin 25^\circ$ is equal to
- A) $\sin 36^\circ$ B) $\cos 36^\circ$ C) $\sin 7^\circ$ D) $\cos 7^\circ$
72. $A_1, A_2, A_3, \dots, A_n$ are n points in a plane whose coordinates are $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$ respectively. A_1A_2 is bisected at the point G_1 , G_1A_3 is divided in the ratio 1:2 at G_2 , G_2A_4 is divided in the ratio 1:3 at G_3 , G_3A_5 is divided in the ratio 1:4 at G_4 , and so on until all n points are exhausted. The coordinates of the final point G so obtained are
- A) $(n(x_1 + x_2 + \dots + x_n), n(y_1 + y_2 + \dots + y_n))$
B) $\left(\frac{x_1 + x_2 + \dots + x_n}{n}, \frac{y_1 + y_2 + \dots + y_n}{n}\right)$
C) $\left(\frac{x_1 + 2x_2 + 3x_3 + \dots + nx_n}{n}, \frac{y_1 + 2y_2 + 3y_3 + \dots + ny_n}{n}\right)$
D) None of these
73. Let PQ and RS be tangents at the extremities of a diameter PR of a circle of radius r . Such that PS and RQ intersect at a point X on the circumference of the circle, then diameter of the circle equals to
- A) $\sqrt{PQ \cdot RS}$ B) $\frac{PQ + RS}{2}$ C) $\frac{2PQ \cdot RS}{PQ + RS}$ D) $\sqrt{\frac{PQ^2 + RS^2}{2}}$

74. The locus of the vertices of the family of parabolas $y = \frac{a^3 x^2}{3} + \frac{a^2 x}{2} - 2a$ is
- A) $xy = \frac{105}{64}$ B) $xy = \frac{3}{4}$ C) $xy = \frac{35}{16}$ D) $xy = \frac{64}{105}$
75. A triangle ABC is placed so that the mid-point of its sides are on the x , y and z axes respectively. Lengths of the intercepts made by the plane containing the triangle on these axes are respectively α, β, γ , then the coordinates of the centroid of the triangle ABC are
- A) $(-\alpha/3, \beta/3, \gamma/3)$ B) $(\alpha/3, -\beta/3, \gamma/3)$
C) $(\alpha/3, \beta/3, -\gamma/3)$ D) $(\alpha/3, \beta/3, \gamma/3)$
76. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{e^{|x|} - e^{-x}}{e^x + e^{-x}}$. Then
- A) f is both one-one and onto B) f is one-one but not onto
C) f is onto but not one-one D) f is neither one-one nor onto
77. The value of $f(0)$ so that the function $f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$ is continuous at each point in its domain, is equal to
- A) 2 B) 1/6 C) 2/3 D) -1/3
78. If $f(x) = (1+x)^n$ ($n \in \mathbb{N}$, $x \in \mathbb{R}$), then the value of $f(0) + f'(0) + \frac{f''(0)}{2!} + \frac{f'''(0)}{3!} + \dots + \frac{f^n(0)}{n!}$ is (where $f^n(0)$ represents n^{th} derivatives of f at 0)
- A) n B) 2^n C) 2^{n-1} D) None of these
79. $\int \frac{dx}{x^{1/2} + x^{1/3}} =$
- A) $2\sqrt{x} - 3x^{1/3} + 6x^{1/6} - \log(1+x^{1/6}) + C$ B) $3x^{1/3} - 6x^{1/6} - \log(1+x^{1/6}) + C$
C) $\sqrt{x} + 3x^{1/3} - 6x^{1/6} + \log(1+x^{1/6}) + C$ D) $3x^{1/3} - \sqrt{x} + 6x^{1/6} - \log(1+x^{1/6}) + C$
80. The solution of the differential equation $y' - Ay = -By^2$ ($A, B > 0$) is (C is arbitrary constant)
- A) $y = ((B/A)e^{Ax} + C)^{-1}$ B) $y = (B/A) + Ce^{-Ax}$
C) $y = e^{-Ax} ((B/A)e^{Ax} + C)^{-1}$ D) $y = e^{Ax} ((B/A)e^{Ax} + C)^{-1}$

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. If $\tan^{-1}\left(x + \frac{2}{x}\right) - \tan^{-1}\left(x - \frac{2}{x}\right) = \tan^{-1} \frac{4}{x}$ then the value of $250x^4 + 320x^2 + 137$ is equal to
82. If O is the origin and the coordinates of A and B are (51, 65) and (75, 81) respectively. Then $OA \times OB \cos \angle AOB$ is equal to
83. Let $0 \leq \beta_r \leq 1$ ($r = 1, 2, \dots$) and $\sum_{r=1}^k \cos \beta_r = \frac{k\pi}{2}$ for any $k \geq 1$ and $A = \sum_{r=1}^k (\beta_r)^r$, then $\lim_{x \rightarrow A} 498 \cdot \frac{(1+x^2)^{1/3} - (1-2x)^{1/4}}{x+x^2}$
84. A positive integer $n \leq 5$ such that $\int_0^1 e^{2x-1} (x-1)^n dx = \frac{7}{4}e^{-1} - \frac{1}{16}(e^2 - 1)$ is
85. Let a, b, c are given vectors, λ, μ, ν are scalars. If $d = \lambda(a \times b) + \mu(b \times c) + \nu(c \times a)$, $[abc] = 1/8$ and $d \cdot (a + b + c) = 8$ then $\lambda + \mu + \nu$ is equal to
86. The maximum value of $f(x) = (\sqrt{4-x^2} - 3)^2 + (\sqrt{4-x^2} + 1)^3$ is
87. Let $\bigcup_{i=1}^{50} X_i = \bigcup_{i=1}^n Y_i = T$, where each X_i contains 10 elements and each Y_i contains 5 elements. If each element of the set T is an element of exactly 20 of sets X_i 's and exactly 6 of sets Y_i 's then n is equal to
88. If $A(4, 3, 5), B(0, 6, 0), C(-8, 1, 4)$ are three consecutive vertices of parallelogram $ABCD$. If θ is the angle between AC and BD the sum of digits of $3025 \sec^2 \theta$ is
89. Let $x_1, x_2, \dots, x_{100}$ be in an arithmetic progression with $x_1 = 2$ and their mean equal to 200. If $y_i = i(x_i - i), 1 \leq i \leq 100$. Let y be the mean of $y_1, y_2, \dots, y_{100}$. Then $y - 9000.5$
90. Let $A = \{1, 2, 3, 4\}$, R be a relation on the set $A \times A$ defined by $R = \{((a, b), (c, d)) : 2a + 3b = 4c + 5d\}$. Then the number of elements in R is